

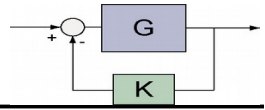
ME 2801

Introduction to Control Systems

W1: Laplace Transform Introduction



Plan of the Day



LTI System Analysis Overview

- Motivation for learning Laplace Transforms

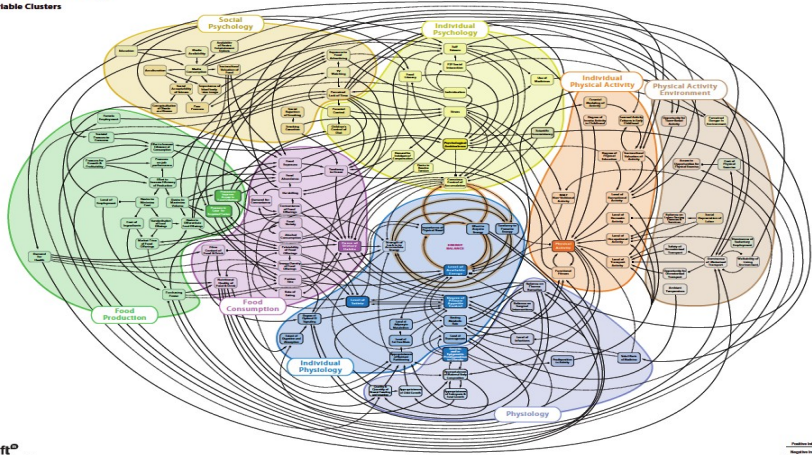
Laplace Transform and Inverse Laplace

- Examples

HW1 Preview

The Art of “Boxes and Arrows”

Obesity System Map
Variable Clusters



Time Domain

Physics:
Newton, Kirchhoff,
1st/2nd laws

Governing Equations (GE)
diff eqn, a set of ODEs

Transforms:
Laplace, Fourier

$\mathcal{L}\{\}$

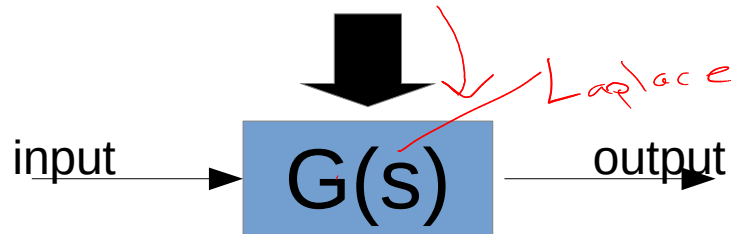
s-Domain

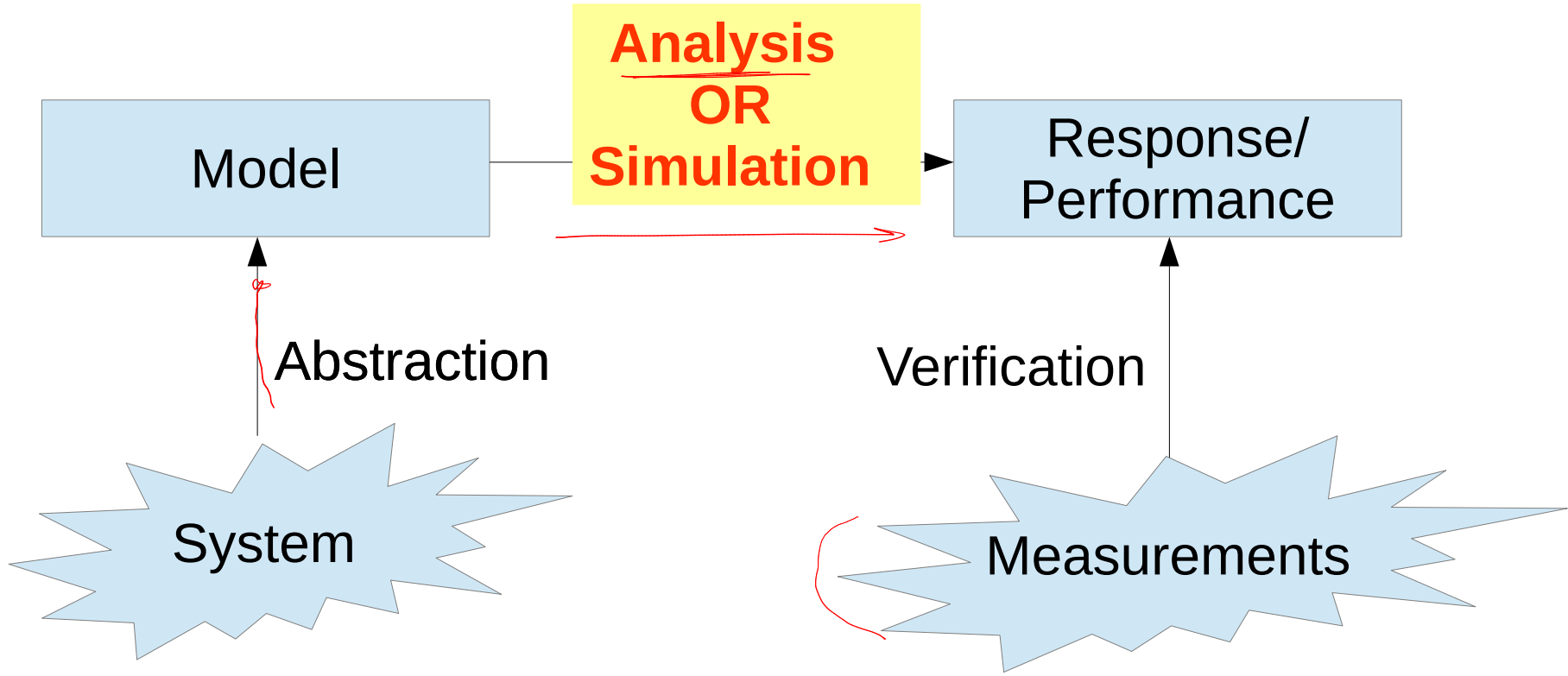
Transfer Function (TF)

$\mathcal{L}^{-1}\{\}$

Time Domain

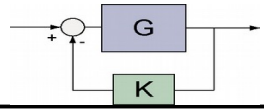
Performance and Stability





Laplace Transforms are a fundamental tool
for rapid system analysis

Laplace Transform



Mathematical tool for representing system models (ODEs).

- Converts differential equations into algebraic equations!

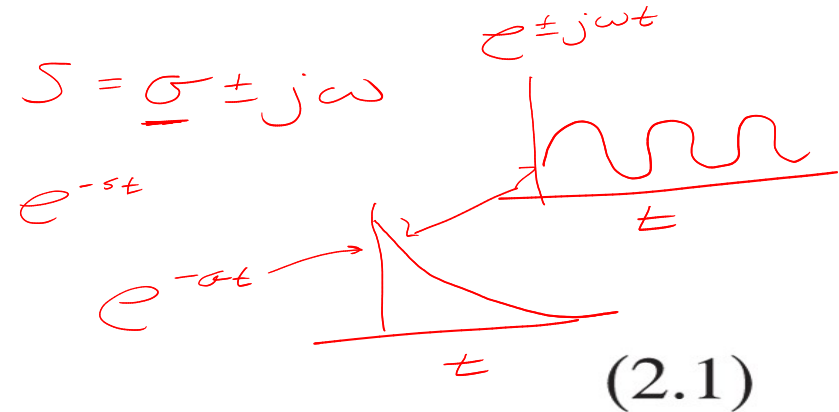
Enables rapid analysis in terms of **inputs** and **outputs**.

Affords design in the **s-domain** with respect to stability and performance.

$$\mathcal{L}[g(t)] = G(s)$$

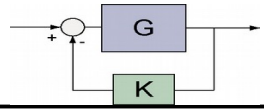
$$s = \sigma \pm j\omega$$

e^{-st}
 $e^{-\sigma t}$
 $e^{\pm j\omega t}$



(2.1)

$$\mathcal{L}[f(t)] = F(s) = \int_{0-}^{\infty} f(t)e^{-st} dt$$



Laplace transform of time function

- $f(t) \rightarrow F(s)$
- Use properties and pairs

Inverse Laplace transform of s-function

- $F(s) \rightarrow f(t)$
- Partial fraction expansion, using MATLAB

Solving ODEs with Laplace

Define a transfer function }

- ODE $\rightarrow G(s)$

"Properties and Pairs"

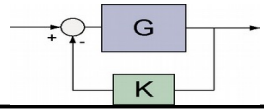


TABLE 2.1 Laplace transform table

Item no.	$f(t)$	$F(s)$
1.	$\delta(t)$	1
2.	$u(t)$	$\frac{1}{s}$
3.	$tu(t)$	$\frac{1}{s^2}$
4.	$t^n u(t)$	$\frac{n!}{s^{n+1}}$
5.	$e^{-at} u(t)$	$\frac{1}{s+a}$
6.	$\sin \omega t u(t)$	$\frac{\omega}{s^2 + \omega^2}$
7.	$\cos \omega t u(t)$	$\frac{s}{s^2 + \omega^2}$

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} f(t) e^{-st} dt$$

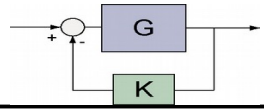
TABLE 2.2 Laplace transform theorems

Item no.	Theorem	Name
1.	$\mathcal{L}[f(t)] = F(s) = \int_0^{\infty} f(t) e^{-st} dt$	Definition
2.	$\mathcal{L}[kf(t)] = kF(s)$	Linearity theorem
3.	$\mathcal{L}[f_1(t) + f_2(t)] = F_1(s) + F_2(s)$	Linearity theorem
4.	$\mathcal{L}[e^{-at} f(t)] = F(s+a)$	Frequency shift theorem
5.	$\mathcal{L}[f(t-T)] = e^{-sT} F(s)$	Time shift theorem
6.	$\mathcal{L}[f(at)] = \frac{1}{a} F\left(\frac{s}{a}\right)$	Scaling theorem
7.	$\mathcal{L}\left[\frac{df}{dt}\right] = sF(s) - f(0-)$	Differentiation theorem
8.	$\mathcal{L}\left[\frac{d^2 f}{dt^2}\right] = s^2 F(s) - sf(0-) - f'(0-)$	Differentiation theorem
9.	$\mathcal{L}\left[\frac{d^n f}{dt^n}\right] = s^n F(s) - \sum_{k=1}^n s^{n-k} f^{(k-1)}(0-)$	Differentiation theorem
10.	$\mathcal{L}\left[\int_0^t f(\tau) d\tau\right] = \frac{F(s)}{s}$	Integration theorem
11.	$f(\infty) = \lim_{s \rightarrow 0} sF(s)$	Final value theorem ¹
12.	$f(0+) = \lim_{s \rightarrow \infty} sF(s)$	Initial value theorem ²

¹For this theorem to yield correct finite results, all roots of the denominator of $F(s)$ must have negative real parts, and no more than one can be at the origin.

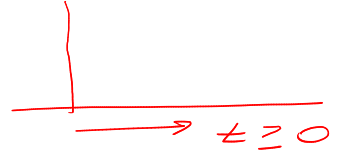
²For this theorem to be valid, $f(t)$ must be continuous or have a step discontinuity at $t = 0$ (that is, no impulses or their derivatives at $t = 0$).

Example: $f(t) \rightarrow F(s)$



Given the following time function, find the Laplace transform.

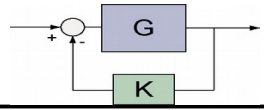
$$f(t) = (1 + 2t)u(t) = \underline{(1 + 2t)u(t)}$$



$$\begin{aligned} \mathcal{L}[f(t)] &= \mathcal{L}[1u(t)] + \mathcal{L}[2tu(t)] \\ &= \frac{1}{s} + 2\mathcal{L}[tu(t)] \end{aligned}$$

$$F(s) = \frac{1}{s} + 2\frac{1}{s^2}$$

Example: $f(t) \rightarrow F(s)$



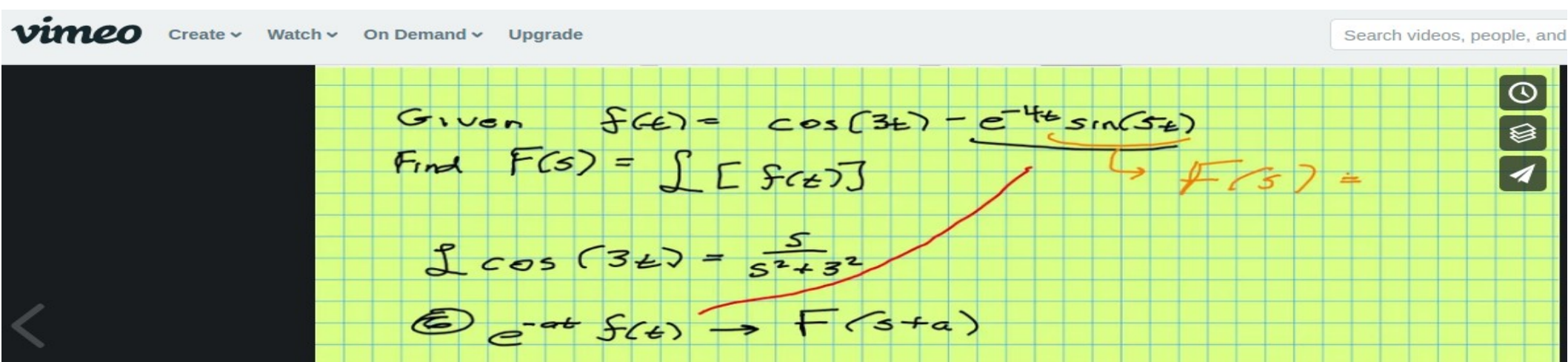
Given the following time function, find the Laplace transform.

$$\mathcal{L}[f(t)] = 5 \cos(7t) + e^{-2t} = F(s)$$

$$F(s) = 5 \left(\frac{s}{s^2 + 7^2} \right) + \frac{1}{s+2}$$

Given the following time function, find the laplace transform.

$$f(t) = \cos(3t) - e^{-4t} \sin(5t)$$



A screenshot of a Vimeo video player showing a handwritten solution on a green grid background. The text in the video is as follows:

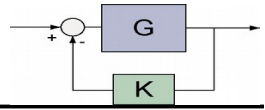
Given $f(t) = \cos(3t) - e^{-4t} \sin(5t)$
Find $F(s) = \mathcal{L}[f(t)]$

$\mathcal{L}[\cos(3t)] = \frac{s}{s^2 + 3^2}$

⑥ $e^{-at} f(t) \rightarrow F(s+a)$

Handwritten notes include an arrow pointing from the e^{-4t} term in the first equation to the $F(s) =$ part, and another arrow pointing from the $\mathcal{L}[\cos(3t)]$ result to the $F(s) =$ part.

Inverse Laplace Transform



Given s-domain function ($F(s)$),
find t-domain function ($f(t)$).

$$\mathcal{L}[f(t)] \rightarrow F(s)$$

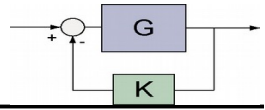
$$f(t) \leftarrow \mathcal{L}^{-1}[F(s)]$$

$$F(s) = \frac{2}{s} + \frac{6}{s^2} + \frac{1}{s+2}$$

$$f(t) = 2\mu(t) + (6t)\mu(t) + e^{-2t}\mu(t)$$

$$= \underline{\underline{(2 + 6t + e^{-2t})\mu(t)}}$$

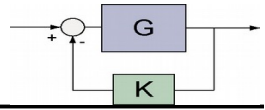
Inverse Lap, : Partial Fraction Expansion



Given s-domain function ($F(s)$),
find t-domain function ($f(t)$).

$$F(s) = \frac{1}{s(s+5)}$$

$$F(s) = \frac{A}{s} + \frac{B}{s+5}$$



Given s-domain function ($F(s)$),
find t-domain function ($f(t)$).

$$F(s) = \frac{1}{s(s+5)}$$

Partial Fraction Expansion

$$F(s) = \frac{1}{s(s+5)} = \frac{A}{\underline{s}} + \frac{B}{\underline{s+5}}$$

Given s-domain function ($F(s)$),
find t-domain function ($f(t)$).

$$F(s) = \frac{1}{s(s+5)}$$

Partial Fraction Expansion

$$F(s) = \frac{1}{s(s+5)} = \frac{A}{s} + \frac{B}{s+5}$$

$s^2 + 5s + 0$

```
>> num = 1; (s+0)(s+5)
den = conv([1 0],[1 5]);
[K,p,k]=residue(num,den)
K =
    -0.2000
     0.2000
p =
    -5
     0
k =
     []
```

Handwritten annotations: -0.2 is linked to the residue at $s = -5$ (labeled $s+5$). 0.2 is linked to the residue at $s = 0$ (labeled s).

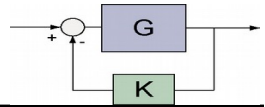
$$F(s) = \frac{1}{s(s+5)} = \frac{-0.2}{s+5} + \frac{0.2}{s}$$

$$f(t) = -0.2e^{-5t} + 0.2u(t)$$

$$= 0.2[1 - e^{-5t}]$$



LTI, ODE Solution w/ Laplace



Given governing equations (LTI, ODE) with initial conditions and input ($u(t)$)

1) Find **Transfer Function** ($G(s)$)

- Transform the equations of motion
 - Apply transform pairs and properties
- Do some algebra to get $G(s)$

2) Find response to input

- Transform $u(t) \rightarrow U(s)$
- Find output via multiplication ($Y(s) = G(s)U(s)$)

3) Find output time response ($y(t)$)

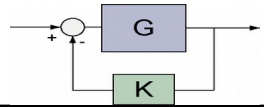
- Inverse Laplace transform $Y(s) \rightarrow y(t)$
 - May require partial fraction expansion



Often Omitted!



HW1 Preview



Given $f(t)$, find $F(s)$

- Properties and pairs

$$\mathcal{L}\{f(t)\} \rightarrow F(s)$$

Create a MATLAB (time-domain) plot ←

Given $F(s)$, find $f(t)$

- Partial fraction expansion (MATLAB: poly and residue)

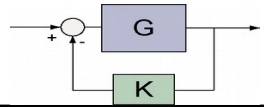
Solving ODEs with Laplace Transforms

- Laplace of ODE
- Solve for output
- Inverse Laplace

Transfer Functions



Next Time



Textbook Reading:

- Chapter 2: 2.1-2.3
- Appendix B (MATLAB): ch2p1-ch2p8 (poly and residue functions)

HW Preview

- Review HW1
- Watch videos on solving ODE's with Laplace
- Come to class with questions